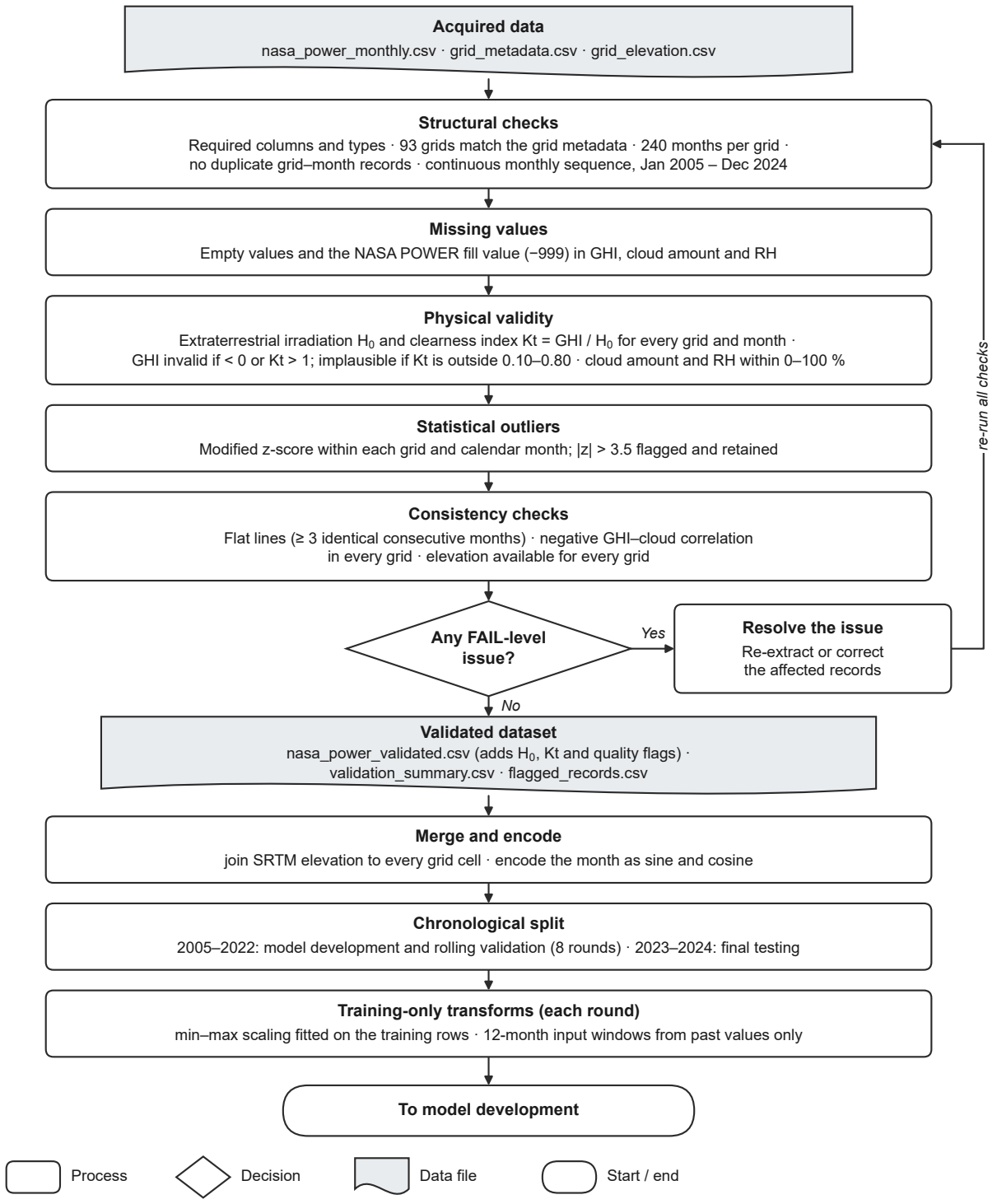

Acquired data
nasa_power_monthly.csv · grid_metadata.csv · grid_elevation.csv
Structural checks
Required columns and types · 93 grids match the grid metadata · 240 months per grid · no duplicate grid–month records · continuous monthly sequence, Jan 2005 – Dec 2024
Missing values
Empty values and the NASA POWER fill value (−999) in GHI, cloud amount and RH
Physical validity
Extraterrestrial irradiation $H_0$ and clearness index Kt = GHI / $H_0$ for every grid and month · GHI invalid if < 0 or Kt > 1; implausible if Kt is outside 0.10–0.80 · cloud amount and RH within 0–100 %
Statistical outliers
Modified z-score within each grid and calendar month; |z| > 3.5 flagged and retained
Consistency checks
Flat lines (≥ 3 identical consecutive months) · negative GHI–cloud correlation in every grid · elevation available for every grid
Any FAIL-level issue?
Yes
Resolve the issue
Re-extract or correct the affected records
re-run all checks
No
Validated dataset
nasa_power_validated.csv (adds $H_0$, Kt and quality flags) · validation_summary.csv · flagged_records.csv
Merge and encode
join SRTM elevation to every grid cell · encode the month as sine and cosine
Chronological split
2005–2022: model development and rolling validation (8 rounds) · 2023–2024: final testing
Training-only transforms (each round)
min–max scaling fitted on the training rows · 12-month input windows from past values only
To model development
Process
Decision
Data file
Start / end